WEEK-4 Assignment

Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

Objective

The objective of this assignment is to understand the basic concepts of Microsoft Azure Cloud and to create a simple data pipeline using Azure Storage Account and Azure Data Factory (ADF). The pipeline reads a CSV file from Azure Blob Storage, checks its metadata, and copies the data to another location.

Task 1: Creating a Resource Group

Procedure

1. Logged in to the Azure Portal using my Azure account.
2. Searched for "Resource Groups" in the search bar.
3. Clicked on "Create Resource Group".
4. Entered the required details such as subscription, resource group name, and region.
5. Clicked on "Review + Create" and then selected "Create".

Result

The resource group was created successfully and will be used to store all resources required for this project.

Task 2: Storage Account and Blob Container Setup

Procedure

Creating a Storage Account

1. Searched for "Storage Accounts".
2. Selected "Create Storage Account".
3. Chose the previously created Resource Group.
4. Entered a unique storage account name.

5. Selected Standard performance and LRS redundancy.
6. Created the storage account.

Creating a Blob Container

1. Opened the created Storage Account.
2. Selected "Containers" under Data Storage.
3. Created a new container named "source-data".

Uploading Dataset

1. Downloaded the Superstore dataset from Kaggle.
2. Opened the Blob container.
3. Clicked on Upload.
4. Selected the Superstore CSV file and uploaded it.

Result

The CSV dataset was uploaded successfully into Azure Blob Storage.

Task 3: Azure Data Factory Basics

Procedure

Creating Azure Data Factory

1. Searched for Azure Data Factory.
2. Clicked on Create.
3. Selected the Resource Group.
4. Entered the Data Factory name.
5. Created the Data Factory instance.

Exploring ADF Studio

After launching ADF Studio, I explored the following sections:

- Author Section – Used for creating datasets and pipelines.

- Monitor Section – Used for checking pipeline execution status.
- Manage Section – Used for configuring linked services and connections.

Creating Linked Service

1. Opened the Manage tab.
2. Selected Linked Services.
3. Added a new Azure Blob Storage Linked Service.
4. Connected it with the Storage Account.
5. Tested the connection successfully.

Creating Datasets

Two datasets were created:

Source Dataset

- Container: source-data
- File: Superstore.csv

Destination Dataset

- Container: processed-data
- File: Processed_Superstore.csv

Using Get Metadata Activity

1. Created a new pipeline.
2. Added the Get Metadata activity.
3. Connected it to the source dataset.
4. Selected metadata fields such as file size, existence, and last modified date.

Result

ADF was successfully connected with Azure Blob Storage and datasets were created.

Task 4: Pipeline Development

Procedure

1. Created a new pipeline named "PL_Copy_Superstore_Data".
2. Added the Get Metadata activity.
3. Added a Copy Data activity.
4. Configured the source dataset as Superstore.csv.
5. Configured the destination dataset as Processed_Superstore.csv.
6. Connected both activities in sequence.

Pipeline Flow:

Get Metadata → Copy Data → Destination File

Result

The pipeline was designed successfully for transferring data from source to destination.

Screenshot

Task 5: Pipeline Execution

Procedure

1. Clicked on Debug to test the pipeline.
2. Alternatively used Trigger Now to execute the pipeline.
3. Monitored the execution status in the Monitor section.
4. Verified the output file in the destination container.

Result

The pipeline executed successfully and the output file was generated in the destination container.

Task 6: IAM Role Assignment

Procedure

Assigning Reader Role

1. Opened Storage Account.
2. Selected Access Control (IAM).
3. Added Reader role to the required user.

Assigning Contributor Role

1. Added Contributor role using IAM settings.
2. Verified the assignment.

Providing Storage Access to ADF

1. Opened IAM settings.
2. Added "Storage Blob Data Contributor" role.
3. Assigned the role to Azure Data Factory Managed Identity.

Result

Required permissions were assigned successfully.

Mini Project: CSV Data Processing using Azure Data Factory

Problem Statement

The task was to create a complete Azure Data Factory pipeline that reads a CSV file from Azure Blob Storage, validates its metadata, and copies it to another location.

Implementation

The Superstore CSV dataset was uploaded into the source Blob container. A Linked Service was created to establish connectivity between Azure Data Factory and Azure Storage. Source and destination datasets were configured. A Get Metadata activity was used to verify the file properties before processing. Finally, the Copy Data activity transferred the file to the destination container.

Output

- Source file successfully detected.
- Metadata validated successfully.
- Data copied to destination container.
- Pipeline execution status showed "Succeeded".

Conclusion

Through this assignment, I learned how Azure resources are created and managed. I understood the working of Azure Blob Storage, Linked Services, Datasets, and Azure Data Factory pipelines. I also learned how to validate files using Get Metadata activity and transfer data using Copy Data activity. The project was completed successfully, and the pipeline executed without errors.